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For Second-year Master2: IA & Data Science

Rapport de TP03: Fire Monitoring Data Analysis and Visualization Using Power BI

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1. Introduction

1.1 General Context

Data analysis and visualization play an important role in modern monitoring systems. They allow raw data to be transformed into meaningful information that supports decision-making and helps users identify patterns, trends, and abnormal conditions.

Fire monitoring is an important application of data analysis. Environmental measurements such as temperature, humidity, and smoke levels can provide useful information about potentially dangerous conditions. Analyzing these measurements can help users understand environmental changes and examine indicators associated with possible fire risks.

Microsoft Power BI is a business intelligence platform that allows users to import, transform, analyze, and visualize data through interactive dashboards. It provides different visual elements, including charts, tables, KPI cards, and filters, making complex datasets easier to understand.

1.2 Artificial Intelligence and Fire Monitoring

Artificial Intelligence (AI) and the Internet of Things (IoT) can be combined to develop intelligent monitoring systems. IoT sensors collect environmental measurements, while data analysis and machine learning techniques can help identify patterns associated with abnormal situations.

In a complete fire-monitoring system, sensors collect data continuously, a detection model analyzes the measurements, and an alarm can be generated when dangerous conditions are identified.

In this practical work, Power BI is used to analyze and visualize simulated fire-monitoring data. The project focuses on presenting environmental measurements and alarm indicators in an interactive dashboard.

The current implementation is a data visualization prototype. It does not include a live sensor connection or a validated real-time fire detection model.

1.3 Objectives

The main objectives of this practical work are:

- To import a fire-monitoring dataset into Power BI.
- To inspect and prepare the data using Power Query.
- To understand the structure and meaning of the dataset variables.
- To create Key Performance Indicators (KPIs).
- To visualize temperature, humidity, smoke levels, and alarm counts.
- To compare measurements across different sensors.
- To create an interactive dashboard using a sensor slicer.
- To interpret the results and discuss the limitations of the project.

2. Dataset Description

2.1 Dataset Overview

The dataset used in this project is a synthetic dataset generated to simulate measurements from a fire-monitoring system.

It contains 10,000 records and seven columns. Each record represents a simulated observation associated with a sensor.

The dataset is stored in CSV format and imported into Power BI Desktop.

Dataset characteristics:

Characteristic	Description
File format	CSV
Number of records	10,000
Number of columns	7
Data type	Synthetic monitoring data
Number of sensor identifiers	10
Analysis tool	Microsoft Power BI Desktop
Application	Fire-monitoring data visualization

2.2 Description of Variables

The dataset contains the following variables:

Variable	Description
`timestamp`	The date and time associated with a measurement.
`sensor_id`	Identifies the sensor associated with a record.
`temperature`	Represents the measured temperature.
`Humidity`	Represents the measured humidity.
`smoke_level`	Represents the simulated smoke level.
`fire_alarm`	Indicates whether a record is labelled as an alarm.
`sensor_status`	Represents the reported status of a sensor.

The `timestamp` variable can be used for time-based analysis, while `sensor_id` allows comparisons between different sensors.

The `temperature`, `humidity`, and `smoke_level` variables describe environmental conditions. The `fire_alarm` variable provides a binary indicator, and `sensor_status` provides categorical information about the sensor.

2.3 Alarm Generation Rule

In the generated dataset, the alarm indicator was created using the following rule:

- If temperature is greater than or equal to 35 and smoke level is greater than or equal to 150, the record is labelled as an alarm.
- Otherwise, the record is labelled as a non-alarm.

The corresponding values are:

- `fire_alarm = 1`: Alarm-labelled record.
- `fire_alarm = 0`: Non-alarm-labelled record.

The dataset contains 750 alarm-labelled records out of 10,000 records.

The alarm percentage is calculated as follows:

Alarm Percentage = (750 / 10,000) × 100 = 7.5%

This percentage describes the generated dataset. It does not represent the real-world probability of a fire.

3. Tools and Methodology

3.1 Tools Used

The project was implemented using the following tools:

Microsoft Power BI Desktop: Used to import the dataset, create visualizations, and build the dashboard.

Power Query Editor: Used to inspect the dataset, review data types, and perform data preparation operations.

CSV Dataset: Used as the source of the simulated fire-monitoring measurements.

Power BI Visualizations: Used to present key indicators, compare sensors, and filter the displayed information interactively.

3.2 Project Workflow

The project follows these main stages:

1. Prepare the synthetic dataset.
2. Import the CSV file into Power BI Desktop.
3. Inspect the dataset using Power Query Editor.
4. Review the column names and data types.
5. Check data quality and apply necessary transformations.
6. Create KPI cards.
7. Build charts for temperature, smoke levels, and alarms.
8. Add a slicer based on sensor identifiers.
9. Analyze the dashboard results.
10. Document the limitations and possible improvements.

4. Data Preparation Using Power Query

4.1 Importing the Dataset

The dataset was imported into Power BI Desktop using the Text/CSV data source option.

The file used in the project is named `tp03_fire_monitoring_sample.csv`.

After selecting the file, the data preview was examined before loading the dataset into Power BI.

This step is important because it allows the user to verify that the expected columns are present and that the data is interpreted correctly.

4.2 Data Inspection

Power Query Editor was used to examine the dataset structure and review its variables.

The following elements are important during data inspection:

- Checking the column names.
- Verifying the data types.
- Identifying missing values.
- Inspecting potential duplicate records.
- Checking for unexpected or inconsistent values.
- Ensuring that numerical variables can be used in calculations.

Data inspection is a necessary step because data quality problems can affect the accuracy of calculations and the interpretation of visualizations.

4.3 Data Types

Appropriate data types are essential for correct analysis.

The recommended data types for the dataset are:

Column	Recommended data type	Purpose
`timestamp`	Date/Time	Time-based analysis
`sensor_id`	Text	Identification and categorization
`temperature`	Decimal number	Average temperature calculations
`Humidity`	Decimal number	Average humidity calculations
`smoke_level`	Decimal or whole number	Smoke-level analysis
`fire_alarm`	Whole number	Alarm counting and summation
`sensor_status`	Text	Categorical analysis

The `sensor_id` column should be treated as an identifier rather than as a numerical measurement. For example, sensor 10 is not necessarily greater in importance or value than sensor 2.

4.4 Data Cleaning and Transformation

Data cleaning helps improve the quality and consistency of a dataset before analysis.

Power Query provides several operations for this purpose, including removing duplicate rows, handling missing values, correcting data types, and renaming columns.

During this stage, the dataset was inspected to determine whether its fields were suitable for use in the dashboard.

Duplicate removal should only be reported as completed after verifying the selected columns and the resulting number of records. Similarly, missing-value removal should only be reported if missing values were actually identified and handled.

After the necessary preparation steps, the data can be loaded into Power BI using the Close & Apply command.

5. Dashboard Design and Implementation

5.1 Dashboard Overview

The dashboard was designed to provide an overview of the simulated fire-monitoring dataset.

It combines numerical indicators and graphical visualizations in a single interface. The dashboard includes:

- Four KPI cards.
- Three horizontal bar charts.
- One clustered column chart.
- One interactive sensor slicer.

The main purpose of the dashboard is to summarize the dataset and allow users to compare measurements and alarm indicators across sensors.

5.2 Key Performance Indicators

Four KPI cards were created to present the main numerical indicators.

KPI	Displayed value	Interpretation
Average Temperature	29.93	Average temperature across the dataset
Average Humidity	52.69	Average humidity across the dataset
Total Alarm Count	750	Total number of alarm-labelled records
Total Records	10,000	Number of records in the dataset

The average temperature is 29.93, while the average humidity is 52.69, according to the current dashboard.

The total alarm count is 750 because the `fire_alarm` column contains 750 records labelled with the value 1.

The total-records KPI represents the number of records included in the current filter context. If a sensor is selected using the slicer, this value may change to reflect the selected sensor.

The units for temperature and humidity should be confirmed from the dataset-generation configuration before interpreting the values physically.

5.3 Average Temperature by Sensor

The first horizontal bar chart displays the average temperature for each sensor.

This visualization allows users to compare temperature measurements across the different sensor identifiers.

It can help identify sensors with higher or lower average temperatures in the generated dataset.

However, a higher average temperature does not necessarily indicate a fire. A real fire-monitoring system would require additional measurements, suitable thresholds, and a validated detection method.

5.4 Average Smoke Level by Sensor

The second horizontal bar chart displays the average smoke level for each sensor.

This chart helps compare simulated smoke measurements across the ten sensors.

Sensors with higher average smoke levels can be examined more closely to understand the distribution of the generated measurements.

Nevertheless, the chart is descriptive. It cannot independently confirm that a fire has occurred or that a sensor is malfunctioning.

5.5 Alarm Count by Sensor

The third horizontal bar chart compares the number of alarm-labelled records associated with each sensor.

This visualization provides an overview of the distribution of alarms in the generated dataset.

It helps users identify which sensors have higher or lower alarm counts and compare their records.

A higher alarm count does not automatically mean that a sensor is located in a more dangerous area. It may depend on the generated measurements and the number of records associated with that sensor.

5.6 Clustered Column Chart

A clustered column chart was created to compare the alarm indicator across sensor identifiers.

This chart presents information about alarm and non-alarm records for each sensor, depending on the aggregation configured in Power BI.

It is important to distinguish between counting the records in the `fire_alarm` column and summing its values.

Counting records includes both zeros and ones, while summing the binary indicator gives the number of alarm-labelled records.

The clustered column chart makes these categories easier to compare visually.

5.7 Interactive Sensor Slicer

A slicer based on the `sensor_id` column was added to the dashboard.

The slicer allows users to select a specific sensor or a group of sensors. The KPI cards and charts can then update according to the selected filter.

This feature improves the usability of the dashboard because users can focus on particular sensors without modifying the original dataset.

The slicer also demonstrates how interactive filtering can support data exploration in Power BI.

6. Results and Discussion

6.1 Summary of Results

The main results obtained from the current dashboard are summarized below.

Indicator	Result
Total number of records	10,000
Total alarm-labelled records	750
Alarm-labelled record percentage	7.5%
Average temperature	29.93
Average humidity	52.69
Number of sensor identifiers	10

These results provide a general overview of the generated monitoring dataset.

6.2 Interpretation of Results

The dataset contains 10,000 records, of which 750 are labelled as alarm records. This corresponds to 7.5% of the dataset.

Because the alarm labels were generated using predefined conditions, this percentage reflects the structure of the synthetic dataset rather than the frequency of real fire incidents.

The average temperature and humidity summarize the environmental variables across all records. However, averages alone cannot describe every observation or identify short periods of unusually high measurements.

The sensor-level charts make it possible to compare temperature, smoke levels, and alarm counts across different sensors. They provide a convenient way to explore the dataset and identify patterns that may require further investigation.

The slicer improves the analysis by allowing users to focus on selected sensors and examine their associated indicators.

6.3 Importance of Interactive Visualization

Interactive dashboards provide several advantages:

- They make large datasets easier to explore.
- They summarize important indicators in a compact interface.
- They facilitate comparisons between sensors.
- They allow users to filter data dynamically.
- They support the interpretation and presentation of results.
- They reduce the need to inspect every record individually.

The dashboard therefore demonstrates how Power BI can be used as a descriptive data analysis tool for simulated environmental monitoring.

7. Geographical Visualization

7.1 Objective

A geographical map could provide an additional view of a fire-monitoring system by displaying sensor locations.

In a real deployment, geographical visualization could help operators identify where sensors are installed and examine the spatial distribution of reported events.

However, a geographical map requires valid location information, such as latitude and longitude coordinates for each sensor.

7.2 Implementation Limitation

During this practical work, an attempt was made to use map-based visualization in Power BI. However, the map visualizations did not render as expected, even after reviewing the map-related settings and geographic field configuration.

As a result, a functional geographical map could not be included in the final dashboard.

The current dataset does not contain verified geographical coordinates for each sensor. Any single location assigned to all records would only serve as a demonstration and would not represent the real position of each sensor.

For these reasons, the dashboard focuses on KPI cards, bar charts, a clustered column chart, and interactive filtering rather than presenting an inaccurate geographical visualization.

7.3 Possible Improvements

Future work could improve geographical analysis by:

- Adding verified latitude and longitude coordinates for each sensor.
- Configuring the geographic fields correctly in Power BI.
- Verifying map visual availability and configuration.
- Using a supported mapping visual with valid location data.
- Displaying sensor locations and alarm indicators on the map.

These improvements would require reliable location data and successful configuration of the mapping component.

8. Limitations of the Project

Although the dashboard provides useful descriptive visualizations, several limitations must be considered.

1. Synthetic dataset: The data was generated for educational purposes. It does not represent verified measurements from a real fire-monitoring deployment.

2. Rule-based alarm labels: The alarm indicator was generated using predefined temperature and smoke-level conditions. Therefore, the dashboard does not demonstrate the accuracy of an independent fire detection model.

3. No live sensor connection: The current implementation analyzes a static CSV dataset rather than continuously receiving measurements from physical sensors.

4. Limited geographical information: The dataset does not provide verified coordinates for individual sensors, and the map visualization was not successfully completed.

5. Descriptive analysis: The dashboard summarizes and compares existing data. It does not independently predict future fires or confirm real emergencies.

6. Data quality verification: Any claim about duplicate removal or missing-value correction must be supported by the actual Power Query operations and the final dataset.

These limitations do not prevent the dashboard from demonstrating the basic principles of data preparation, analysis, and interactive visualization.

9. Future Improvements

Several improvements could extend this project.

9.1 Integration with Real Sensors

The system could be connected to physical sensors capable of measuring temperature, humidity, and smoke-related variables.

Measurements could be transmitted periodically to a database or data-processing service, allowing the dashboard to display updated information.

9.2 Integration of Machine Learning

A machine learning model could be trained and evaluated using an appropriate labelled dataset.

Its predictions could then be compared with actual observations and, if suitable, incorporated into the monitoring workflow.

Before deployment, the model would need to be evaluated using relevant metrics, such as precision, recall, F1-score, and false-alarm rate.

9.3 Real-Time Monitoring

A future version could integrate a streaming system or message broker to receive sensor measurements continuously.

The data could be processed and stored before being presented through a monitoring interface.

9.4 Improved Geographical Visualization

Verified coordinates could be associated with each sensor to support geographical analysis.

A functioning map could then display sensor locations and, where appropriate, the status of individual sensors.

9.5 Additional Analytical Indicators

The dashboard could be extended with:

- Alarm percentages by sensor.
- Temperature and smoke-level trends over time.
- Daily and hourly alarm distributions.
- Sensor status summaries.
- Historical comparisons.
- Alerts based on clearly defined and validated criteria.

These additions could make the dashboard more informative and suitable for more advanced analysis.

10. Conclusion

This practical work focused on the analysis and visualization of simulated fire-monitoring data using Microsoft Power BI.

A dataset containing 10,000 records and seven variables was used to build an interactive dashboard. The dashboard includes four KPI cards showing average temperature, average humidity, total alarm count, and total number of records. It also includes three horizontal bar charts, a clustered column chart, and a sensor slicer.

The results show 750 alarm-labelled records, corresponding to 7.5% of the generated dataset. The visualizations make it easier to compare environmental measurements and alarm distributions across the ten sensor identifiers.

The practical work also demonstrated the importance of data inspection, appropriate data types, and interactive filtering. Although geographical visualization was attempted, the map could not be successfully integrated into

Overall, this project demonstrates the usefulness of Power BI for descriptive analysis and interactive visualization. It also provides a foundation for future development involving real sensor data, validated machine learning models, real-time monitoring, and reliable geographical information.

It is important to emphasize that the current dashboard is an educational prototype based on synthetic data. It should not be considered a validated fire detection or emergency response system.

the final dashboard.

